

Learning Resources — Python for Beginners

M2 EMN Online Neurosciences 2026–2027 — Introduction to Programming

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Use the notebook [Scripts/Python_intro_neuroscience_EN.ipynb](#) (see [tutorial_python_intro_neuroscience_notebook.md](#)) as your starting point, then pick one or two resources below depending on whether you prefer reading, watching, or doing interactive exercises.

1. Interactive courses (learn by doing)

These platforms let you write and run code directly in the browser — good for building muscle memory early on.

- **DataCamp** (*English*) — Structured, beginner-friendly courses with in-browser coding exercises. Look for the **"Introduction to Python"** and **"Intermediate Python"** courses in their Python track. Free courses are limited, but their intro content is a good, guided starting point and is widely used in data-science / scientific-computing curricula (relevant for the NumPy/Matplotlib parts of our notebook).
- **Codecademy — Learn Python 3** (*English*) — Interactive, step-by-step course, free tier available.
- **Kaggle Learn — Python** (*English*) — Short, free, hands-on course, useful once you want to move toward data analysis.
- **Sololearn** (*English*) — Bite-sized interactive lessons, good for reviewing concepts on a phone between sessions.

2. Free written tutorials / documentation

- **The Official Python Tutorial** (*English*) — Written by the Python developers themselves. Dense, but authoritative and always up to date. Good as a reference once you already know the basics.
- **Automate the Boring Stuff with Python** (*English, free online book*) — Extremely popular with true beginners; explains concepts with very practical, everyday examples (files, spreadsheets, etc.).
- **Real Python** (*English*) — Large library of tutorials ranging from beginner to advanced, including many on NumPy and Matplotlib.
- **OpenClassrooms** (*French*) — Search for their **"Apprenez à programmer en Python"** course; clear, structured, and free to read.

3. YouTube channels

French:

- **Machine Learnia** — Made by Guillaume Saint-Cirgue, this is one of the most popular French channels for learning Python, data science, and machine learning from scratch. Look for his **"Apprendre Python"** playlist for complete beginners — very clear, example-driven explanations, ideal as a French-language complement to this course.

English:

- [freeCodeCamp.org](#) — Publishes full-length (several hours) free Python courses for complete beginners, great for a deep, one-session immersion.
 - [Corey Schafer](#) — Very clear, well-paced tutorials on Python fundamentals as well as NumPy, Matplotlib, and Jupyter notebooks specifically.
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4. How to use these resources alongside this course

- If you prefer **structured, interactive practice** → start with DataCamp or Codecademy.
- If you prefer **reading at your own pace** → Automate the Boring Stuff or Real Python.
- If you prefer **watching and following along** → Machine Learnia (French) or freeCodeCamp/Corey Schafer (English).
- Once you're comfortable with the basics, revisit the notebooks from this course (Nernst/GHK, and later Hodgkin-Huxley) — applying new tricks you pick up from these resources to code you already understand is one of the fastest ways to consolidate learning.

If you have questions or want a recommendation tailored to your background, contact:

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